ANSWER KEY & MARKING SCHEME SET A

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 12	neutrons = 23 − 11
2	**(a)** 30	neutrons = 56 − 26
3	**(a)** 2, 8, 6	fill shells in order with maximum 2, 8, 8, 2
4	**(c)** 2	electrons lost, gained or shared to complete the octet
5	**(d)** 18	$2n^2 = 2 \times 3^2 = 18$
6	**(b)** Leucippus and Democritus	ancient Greece
7	**(d)** the positive charge is concentrated in a tiny dense nucleus	nuclear model
8	**(c)** James Chadwick	1932
9	**(c)** protons and neutrons	nucleons
10	**(a)** carbon-14	radiocarbon dating

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Acharya Kanada (parmanus) and Leucippus/Democritus (atomos) (1+1).	1+1
12	Cathode rays were the same irrespective of the cathode material or the gas used, so electrons are a basic component of all atoms (2).	2
13	An atom is a sphere of positive charge in which negatively charged electrons are embedded, like plums in a pudding; overall neutral (2).	1+1
14	Most passed straight through (1); most of the atom is empty space (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Kanada's parmanus and Democritus' atomos were imagined indivisible particles (1.5); Dalton's atomic theory (1808) was based on experiments (1.5).	1½+1½
16	Gas at low pressure in a tube with two electrodes and high voltage; rays travelled from cathode to anode (1); bent by electric and magnetic fields so they are negatively charged particles with mass much smaller than atoms (1); present in every element (1).	1+1+1
17	It predicted that α-particles would pass through or be deflected slightly (1); observed: some deflected sharply, a few rebounded (1); positive charge is concentrated, not spread (1).	1+1+1
	OR: α-particles aimed at a very thin gold foil (1); most passed undeflected, some deflected through small angles, very few rebounded (2).	1+2
18	Positive charge and most mass concentrated in a tiny nucleus; electrons revolve around it like planets; most of the atom is empty (1+1+1).	1+1+1
	OR: $10^{-10}/10^{-15} = 10^5$ times (1); if the atom were a 100 m cricket ground, the nucleus would be a black pepper grain (1) at the centre (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Gas at low pressure in a glass tube with two electrodes and high voltage; rays from cathode deflected in electric and magnetic fields (2). (ii) Cathode rays are streams of negatively charged electrons, independent of the cathode material and the gas, so electrons are present in all atoms (1). (iii) A positive sphere with embedded electrons; atoms are neutral (2).	2+1+2
	OR: (i) α-particles on a thin foil: most undeflected, some slightly deflected, a few rebounded (2). (ii) Atom is mostly empty; positive charge and mass concentrated in a small nucleus; electrons revolve around (2). (iii) More deflections and fewer particles passing through (1).	2+2+1

20	(i) About 10^5 times (1). (ii) The nucleus (1). (iii) Most of the atom is empty space (1). (iv) Accelerating electrons would lose energy and spiral into the nucleus (2).	1+1+1+2
	OR: (i) Electrons move in fixed shells (energy levels) without losing energy; shells farther out have higher energy; transitions need fixed energy; each shell has a limited capacity (2). (ii) In stationary states the energy of the electron stays constant (1). (iii) From Barkla's X-ray series naming — K was the first observed; Bohr adopted it (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET B

Chapter 4: **Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 18	neutrons = 35 − 17
2	**(a)** 6	neutrons = 12 − 6
3	**(c)** 2, 8, 8, 1	fill shells in order with maximum 2, 8, 8, 2
4	**(b)** 3	electrons lost, gained or shared to complete the octet
5	**(a)** 8	$2n^2 = 2 \times 2^2 = 8$
6	**(c)** streams of negatively charged particles	electrons
7	**(c)** 10^5 (one lakh)	10^{-10} m versus 10^{-15} m
8	**(b)** electron	about 1/1836 of a proton
9	**(d)** $^{35}_{17}Cl$ and $^{37}_{17}Cl$	same Z, different A
10	**(d)** number of electrons and electronic configuration	chemical properties depend on valence electrons

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The nucleus is small, dense and positively charged containing most of the mass; electrons revolve around it; most of the atom is empty (1+1).	1+1
12	It could not explain the stability of the atom — an orbiting (accelerating) electron should lose energy and spiral into the nucleus (2).	2
13	Electrons revolve in fixed shells/energy levels without losing energy; they move between shells by absorbing/releasing fixed energy; shells hold a fixed number of electrons (1+1).	1+1
14	A helium atom has 2 protons but mass about 4 times that of hydrogen, so uncharged particles must be present (1); Chadwick discovered the neutron (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$0.1\ mm = 10^{-4}$ m (1); atoms $= 10^{-4}/10^{-10}$ (1) $= 10^6$ (about a million) (1).	1+1+1
16	Electrons move in fixed shells called stationary states and do not lose energy while in them (1.5); so atoms are stable; electrons jump between shells by absorbing or emitting fixed energy (1.5).	1½+1½
17	Charles Barkla named the first observed X-ray series K to leave room for series discovered earlier (1.5); Bohr adopted the same notation (1.5).	1½+1½
	OR: He found a neutral particle of mass about that of a proton (1); He: 2 p + 2 n; mass ≈ 4 u (1); neutrons reduce proton–proton repulsion and add to the nuclear force (1).	1+1+1
18	Berzelius proposed using Latin names for international use (1); Fe (ferrum), Na (natrium), Hg (hydrargyrum) (1); first letter capital, second small (1).	1+1+1
	OR: Neutrons 20, electrons 20 (1); configuration 2, 8, 8, 2 (1); valency 2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Mass of helium is twice what protons alone give (1). (ii) James Chadwick in 1932 discovered the neutral neutron (1). (iii) Protons and neutrons (1). (iv) Neutrons reduce repulsion between protons and strengthen the nuclear force (2).	1+1+1+2
	OR: (i) Maximum $2n^2$ per shell; outermost shell ≤ 8; filled stepwise K, L, M, N (2). (ii) 2,8,2; 2,8,7; 2,8,6; 2,8,8,1; 2,8,8,2 (2.5). (iii) 2, 1, 2, 1, 2 (½).	2+2½+½
20	(i) 17 p, 18 n, 17 e (1). (ii) 2, 8, 7 (1). (iii) Valency 1; chlorine (1). (iv) It gains one electron to form Cl^- (anion) (2).	1+1+1+2

	OR: (i) Valency: number of electrons gained, lost or shared to complete octet; valence electrons: electrons in the outermost shell (2). (ii) Na: loses 1 (valency 1); O: gains 2 (valency 2); C: shares 4 (valency 4) (2). (iii) They have complete valence shells (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND